

# ANH LE

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## PROJECTS

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### Foundation Model Adaptation for *Thin-Structure* Segmentation

Hackathon challenge, team of 3 | gallery

Fargo, ND  
Jul. 2026 – Present

- Benchmarked six segmentation architectures spanning from-scratch CNNs to foundation-model encoders: pretraining delivered **efficiency, not accuracy** (0.627 vs. 0.624 Dice score) at **7× fewer** trained weights (5.4M vs. 37M).
- Fine-tuned that **SAM 2.1** (Hiera-B+) encoder using **LoRA** in a low-data regime (**958** solar images), achieving **0.627 Dice** (vs. 0.596 for full fine-tuning of the same encoder) while updating under **1%** of encoder weights.
- Continued **self-supervised domain-adaptive pretraining** (masked image modeling) of the same encoder on **100k unlabeled** solar images, then re-ran the identical LoRA fine-tune to measure the effect (0.61 vs. 0.63 Dice).

### Adapting a DINOv3 Foundation Model to Seismic Fault Segmentation

Self-directed research study | gallery

Fargo, ND  
Aug. 2026 – Present

- Adapted a **DINOv3** ViT-B/16 encoder to seismic fault segmentation, comparing four levels of fine-tuning (frozen, LayerNorm-only, LoRA, and full), which raised IoU score on synthetic data from 0.613 to 0.707.
- Showed this ordering inverts on real field data (CRACKS): the lightest adaptation, tuning only the encoder's Layer-Norm parameters, beat full fine-tuning (IoU 0.125 vs. 0.109).
- Trained on synthetic data only, then tested on real seismic sections: IoU fell from 0.66 to 0.11, still above the 0.04 agreement between the expert and a practitioner annotator.

### Zero-Shot Video Object Segmentation and Tracking on UAV Footage

Manuscript in progress | gallery

Fargo, ND  
Oct. 2024 – Present

- Matched a ground-truth field sensor within **1%** using **SAM 2.1 Hiera-Large** with no training data, by prompting it for masks of the fixed ground.
- Removed drone drift by matching fixed points inside those masks across frames, dropping bad matches, and solving one correction per frame.
- Ran a second pass of the same model as the tracker, propagating one mask per floating object so centroid displacement became a velocity measurement.

### YourEyes: Vision-Based Indoor Navigation for Blind Users

iOS application, team of 3 | gallery

Fargo, ND  
Jun. 2026 – Present

- Embedded indoor photos with frozen **DINOv2** and benchmarked retrieval accuracy against open-source baselines.
- Built a backend service performing vector **similarity search** over the embedding database.
- Built photo-burst capture carrying heading and altitude metadata, with spoken turn-by-turn guidance.
- Ran on-device LiDAR depth sensing to raise obstacle alerts inside the same real-time capture loop.

## RESEARCH EXPERIENCE

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### Graduate Research Assistant — ICE Lab & AI SUSTEIN, NDSU

Bathymetry Inversion from UAV Video

Fargo, ND  
Oct. 2024 – Present

- Reconstructed bed elevation with  $R^2$  of **0.85** against the surveyed bed profile, using only surface velocity derived from the frozen model's masks (**SAM 2.1 Hiera-Large**).
- Built the full path from raw video to physical result, so a general-purpose vision model, not a site-specific one, replaced a boat-based survey.
- Fed that velocity into a physics-constrained network that enforces the shallow water equations, with measured flow rate as a soft constraint.

## EDUCATION

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### Ph.D. Civil & Environmental Engineering

North Dakota State University (NDSU), Fargo ND

*Expected Dec 2027*

### M.S. & B.S. Civil Engineering, HCMUT, Vietnam

*2020 / 2018*

## TECHNICAL SKILLS

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- **Languages:** Python, MATLAB, SQL, Bash
- **Deep learning:** PyTorch, JAX, Hugging Face Transformers
- **Imaging & CV:** OpenCV, scikit-image, PIVlab, SAM, DINO
- **Data & infrastructure:** PostgreSQL + pgvector, Docker, Git, SLURM